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ಎಐಸಿಟಿ ನವದೆಹಲಿ ಇಂದ ಮಾನ್ಯತೆ ಪಡೆದ - ಫಿಟಿಯು ಬೆಳಗಾವಿ ಸಂಯೋಜಿತ ಸ್ವಾಯತ್ತ ಸಂಸ್ಥೆ

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Major Project-1

An Advanced Intelligent Framework for Personalized Diet Planning, Calorie Tracking, and Health Monitoring using SmartDiet+

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Abstract

SmartDiet+ is an AI-powered dietary assistant that provides personalized meal plans and smart meal suggestions based on user inputs such as age, gender, city, occupation, diet preferences, and allergies. It offers multiple healthy options for breakfast, lunch, and dinner using easily available ingredients, while filtering out allergenic items. The system includes integrated calorie tracking, BMI calculation, and in-built recipe viewing to support user engagement and healthy eating habits. Developed using Flask, HTML, CSS, and JavaScript, SmartDiet+ also features visual tools and optional professional nutritionist guidance to deliver a complete, user-friendly wellness solution.

Contents

1. Introduction
2. Objective
3. Problem Statement
4. Literature Survey
5. Requirement Analysis
6. Proposed Design/Methodology
7. Conclusion
8. Reference

1.Introduction

.Maintaining a well-balanced diet is crucial for overall health, yet many individuals struggle with effectively tracking daily calorie intake and planning meals. Traditional diet planning often lacks personalization, proving ineffective for users with specific health goals, dietary restrictions, or lifestyle preferences. Manual calorie tracking is often inaccurate, time-consuming, and prone to errors. These challenges are exacerbated by busy work schedules, a lack of nutritional knowledge, and limited time for meal preparation, contributing to unhealthy eating habits and a rise in lifestyle-related health issues such as obesity, type 2 diabetes, and heart disease, all directly linked to poor nutrition and severely impacting quality of life.

To address these critical limitations, this project aims to develop an intelligent diet recommendation system that offers personalized meal plans, automates calorie tracking, and provides smart meal planning based on individual preferences and health requirements. By leveraging web technologies and AI-driven algorithms, the system will simplify nutrition management, helping users achieve their dietary goals more effectively and effortlessly.

The Need for Smart Diet Systems

Rising Obesity Rates: Obesity is a global epidemic, affecting over 650 million adults. This underscores the urgent need for effective dietary management tools.

Diet-Related Diseases: Diseases like heart disease and diabetes incur significant costs, reaching ₹20.88 trillion annually. Poor nutrition is directly linked to these conditions, significantly impacting quality of life.

Limitations of Traditional Dieting: Conventional methods often lack personalization and adherence, failing to consider crucial individual factors such as age, gender, daily activity level, food allergies, or cultural eating habits. This project offers a smarter, adaptive approach to dieting, addressing these critical limitations.



2.Objectives

- **Personalized Nutrition Guidance:** Develop an intelligent system that generates customized meal plans based on the user's comprehensive health profile, specific dietary restrictions, and defined fitness goals. This includes matching diet preferences with available meal plans and considering local food availability (optional).
- **Automated Calorie Tracking:** Integrate a robust nutrition database and food recognition APIs to automate calorie calculations, thereby significantly reducing manual effort and improving accuracy.
- **Smart Meal Planning:** Implement an AI-driven meal planner that suggests optimized meal combinations, considering available ingredients (via user input or grocery list integration) and daily nutritional requirements. This system will also allow users to dynamically modify meal plans based on preferences.
- **Health Monitoring & Diet Optimization:** Continuously analyze user data to provide real-time diet suggestions and adaptive meal plans that evolve with user progress and changing goals. This includes refining recommendations based on user behavior and suggesting alternatives if meals are skipped or modified frequently.
- **Food Recommendation System:** Beyond planning, the system will actively suggest healthier alternatives when users log high-calorie or unbalanced meals, guiding them towards better choices.
- **User Progress Tracking:** Enable users to effectively monitor their dietary habits through an interactive dashboard that displays nutritional trends, caloric balance, and adherence to meal plans.

3.Problem Statement

The Growing Challenge of Personalized Nutrition

In today's busy and fast-moving world, many people struggle to maintain a healthy diet. Busy work schedules, lack of nutritional knowledge, and limited time for meal planning often lead to unhealthy eating habits. This problem is becoming more serious as we see a growing number of lifestyle-related health issues such as obesity, type 2 diabetes, and heart diseases. These conditions are directly linked to poor nutrition and can significantly affect a person's quality of life.

Traditional diet plans often fail to meet individual needs. They are usually one-size-fits-all and don't consider important personal factors like age, gender, daily activity level, food allergies, or cultural and regional eating habits. For example, a working adult in a metropolitan city may have very different dietary requirements compared to a student living in a small town, but most diet plans don't reflect these differences.

Moreover, keeping track of daily calorie intake, nutrients, and meal balance manually can be overwhelming and time-consuming. It's easy to make mistakes or lose motivation, which makes it hard for people to stay consistent. Although artificial intelligence (AI) has made great progress in other fields, its integration into personal nutrition and fitness planning is still limited. There is a growing need for smart systems that can provide real-time, personalized advice—adapting to changing goals, preferences, and lifestyles—to help individuals make healthier choices without added stress or confusion.

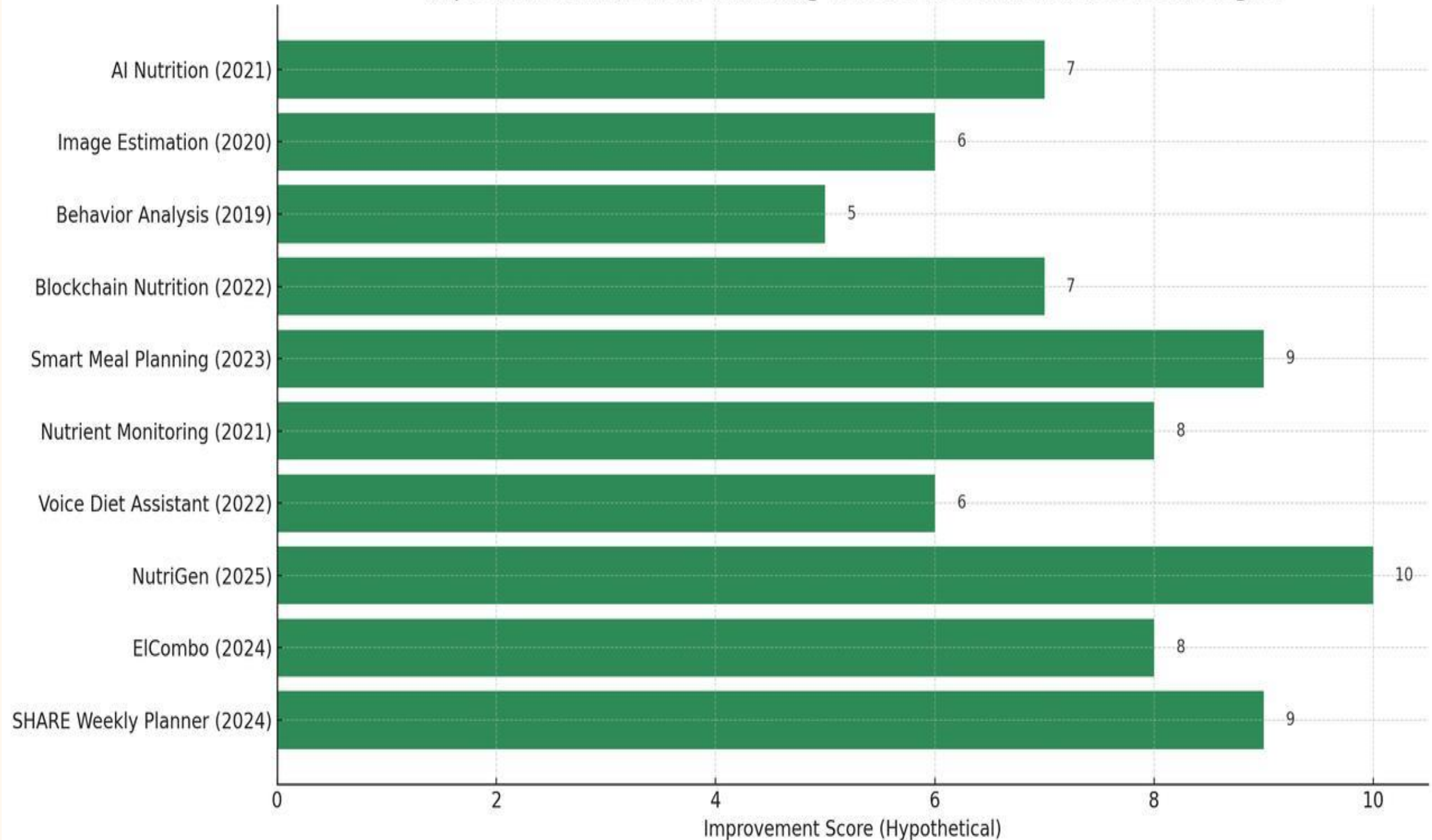
4. Literature Survey

Comparison of Existing Research Studies on Diet Recommendation Systems

No.	Study	Technology Used	Research Focus	Applications	Limitations
1	AI-Based Personalized Nutrition System (2021)	Machine Learning (ML)	Personalized meal recommendations based on past food habits	AI-driven diet coaching apps	Requires large training data, slow adaptation
2	Calorie Estimation Using Image Recognition (2020)	CNN (Computer Vision)	Estimating calorie intake using food image analysis	Mobile calorie tracking apps	Accuracy depends on image quality and food database
3	Dietary Tracking and Behavioral Analysis (2019)	Data Analytics	Long-term tracking of eating habits	Habit-forming diet apps	Requires continuous manual input
4	Nutrition Recommendation System Using Blockchain (2022)	Blockchain	Secure diet tracking with tamper-proof records	Medical diet tracking systems	High computational cost
5	AI-Enhanced Smart Meal Planning (2023)	AI-based meal planning optimization	Adjusting meal plans based on food availability and calories	Grocery and meal kit delivery apps	Needs constant updates to food databases

No.	Study	Technology Used	Research Focus	Applications	Limitations
6	Smart Nutrient Intake Monitoring (2021)	IoT & Wearable Devices	Tracking nutrients through smart sensors	Integration with fitness trackers and smartwatches	High cost of wearable devices
7	Voice-Activated Diet Assistant (2022)	NLP (Natural Language Processing)	Voice-based diet logging and recommendations	AI-powered virtual assistants like Alexa & Google Assistant	Limited voice recognition accuracy for complex meal descriptions
8	NutriGen: Personalized Meal Plan Generator Leveraging Large Language Models (2025)	Large Language Models (LLMs)	Generating personalized meal plans aligning with user-defined dietary preferences and constraints	AI-driven personalized nutrition planning	Requires extensive user data and computational resources
9	ElCombo: Personalized Meal Recommendation System for Chinese Older Adults (2024)	Knowledge Graphs, Personalized Algorithms	Providing meal recommendations considering disease-related nutritional constraints and personal dietary preferences	Enhancing dietary quality among older adults	Limited to specific demographic and cultural dietary data
10	SHARE: Healthy Personalized Recipe Recommendations for Weekly Meal Planning (2024)	Collaborative Filtering, Content-Based Filtering	Offering comprehensive meal plans considering individual dietary tastes, health histories, and dietary restrictions	Personalized weekly meal planning applications	May require continuous user feedback for optimal performance

Improvement in Health Monitoring via Diet Recommendation Technologies





5. Requirement Analysis

Functional Requirements:

- User Registration and Profile Management (age, weight, dietary preferences).
- Personalized Meal Planning (customized diet suggestions).
- Calorie Tracking System (automated intake monitoring).
- Smart Meal Planner (meal suggestions based on available ingredients).

Non-Functional Requirements:

- **Scalability:** Should handle multiple users efficiently.
- **Security:** Protect user data using secure authentication.
- **Usability:** Ensure an intuitive UI with responsive design.

Technology Stack:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python (Flask/Django) or Node.js
- **Database:** MySQL or Firebase
- **API Integration:** Nutrition APIs (for food data and calorie counts)

6. Proposed Design / Methodology

The Diet Recommendation System will be developed using a modular approach, ensuring flexibility, scalability, and adaptability. The system consists of the following core components:

1.User Profile Setup

- Users enter basic details (age, weight, height, activity level, dietary preferences, and health goals).
- The system calculates Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE) to determine personalized calorie requirements.

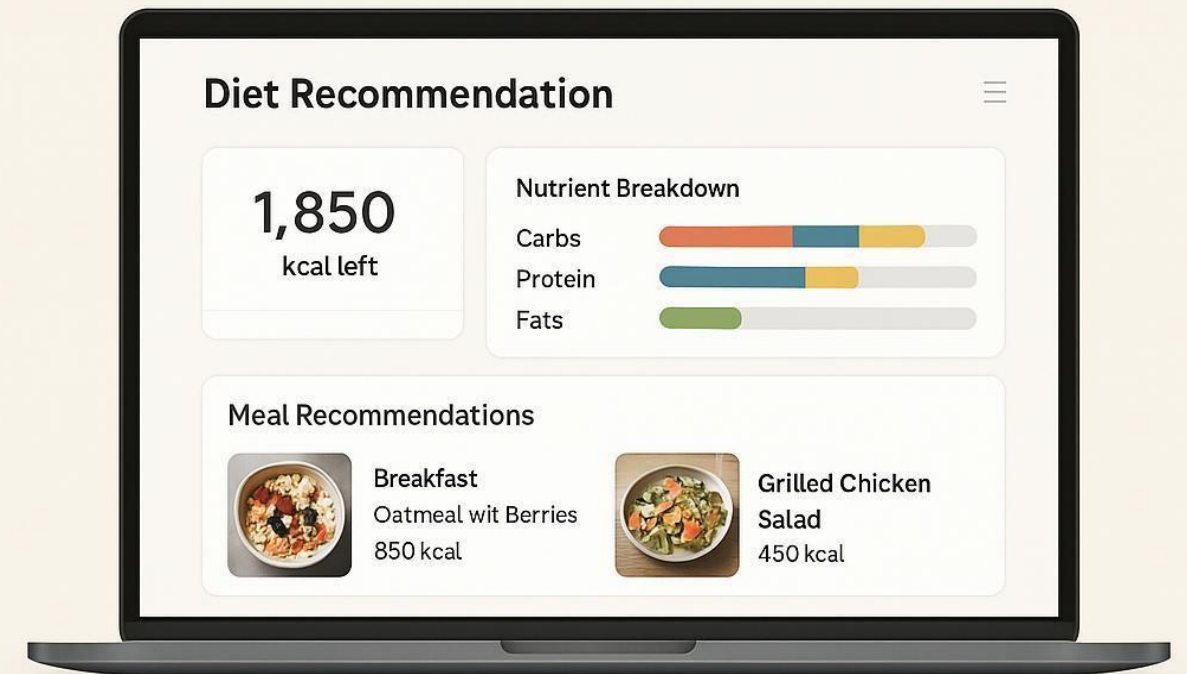
2.AI-Powered Diet Recommendation Engine

- Uses machine learning models to analyze user input and suggest customized meal plans based on macronutrient balance (protein, carbs, fats).
- Integrates with nutrition APIs to fetch calorie and nutrient values of foods.

3.Smart Meal Planning System

- Suggests meals based on available ingredients in the user's kitchen (via user input or grocery list integration).
- Users can modify meal plans dynamically based on preferences.

5. Proposed Design / Methodology



Source: OpenAI

4.Automated Calorie Tracking

- Integrates food recognition APIs (e.g., FoodData Central, Edamam) to estimate calories.
- Allows manual food logging for items not found in the database

5.User Dashboard & Health Insights

- Displays daily calorie intake, nutrient breakdown, and meal plan adherence.
- Provides dietary trend analysis and personalized health tips.

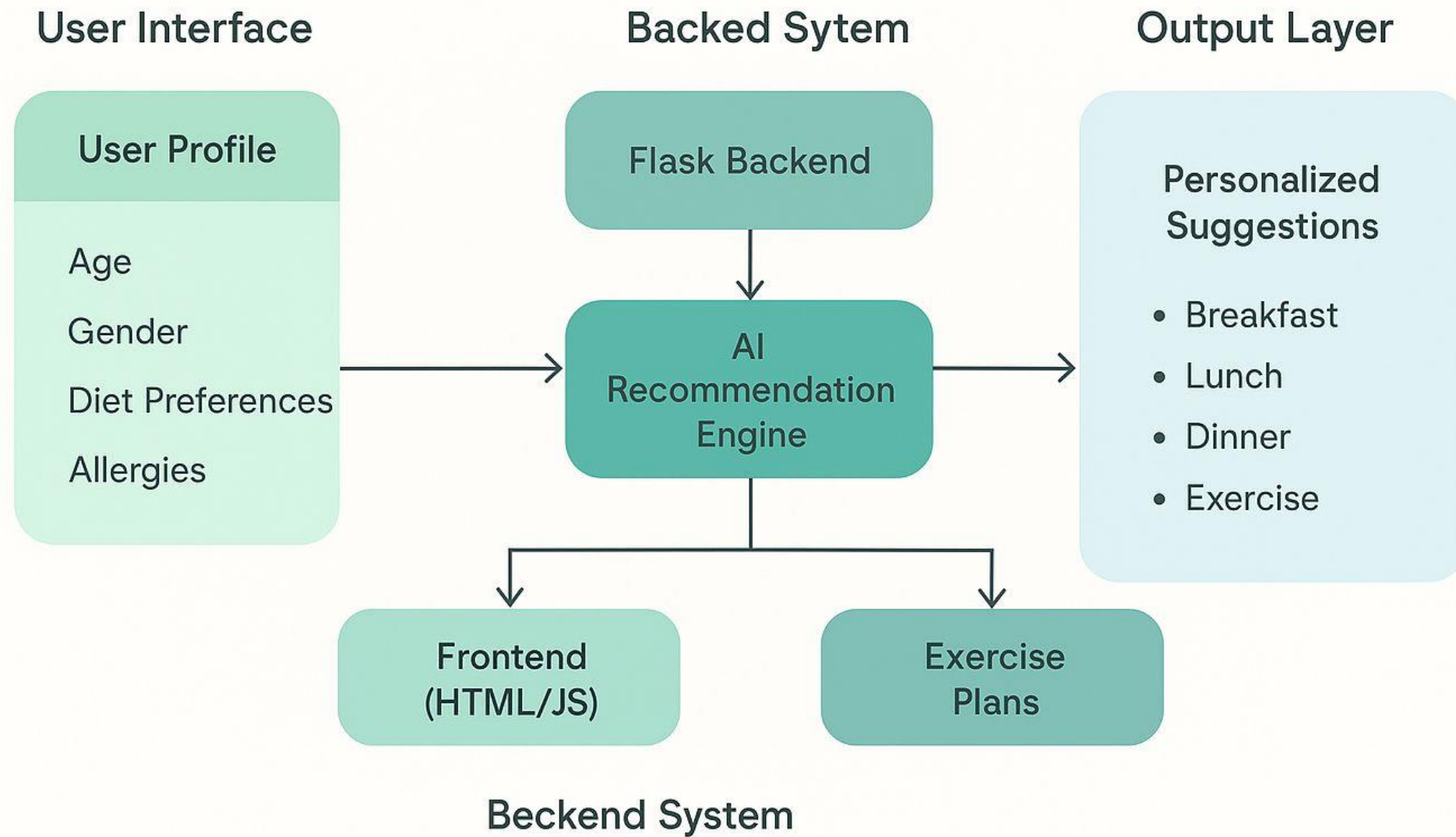
6.Adaptive Learning & Feedback System

- Uses AI-based feedback loops to refine meal recommendations based on user behavior.
- Suggests alternative meals if a user skips or modifies their diet frequently.

7.Cross-Platform Integration

- Web-based UI using HTML, CSS, JavaScript.
- Backend with Python (Flask/Django) or Node.js.
- Integration with mobile apps and wearable devices for seamless calorie tracking.

System Architecture



Algorithm: Smart Diet+ Recommendation System

Step 1: Start

Step 2: Collect user input from the web form:→ Age, Gender, Occupation, Diet Preference, City, Allergies

Step 3: Send the input data to the Flask backend via POST request (/recommend route)

Step 4: In the backend:

a. Validate input data

b. Normalize data (e.g., lowercase diet preferences, trim whitespace)

Step 5: Analyze user profile using logic or a lightweight AI model: Check age group for calorie needs Match diet preference with available meal plans Consider city for local food availability (optional) Remove any options that include allergens

Step 6: Select a suitable meal recommendation from the dataset or logic rules

Step 7: Select a matching exercise plan based on Age Occupation (e.g., sedentary vs active) User goals (optional: e.g., weight loss, maintenance)

Step 8: Prepare final response (Smart Meal + Smart Exercise Plan)

Step 9: Return the result to the frontend and display it to the user

Step 10: End

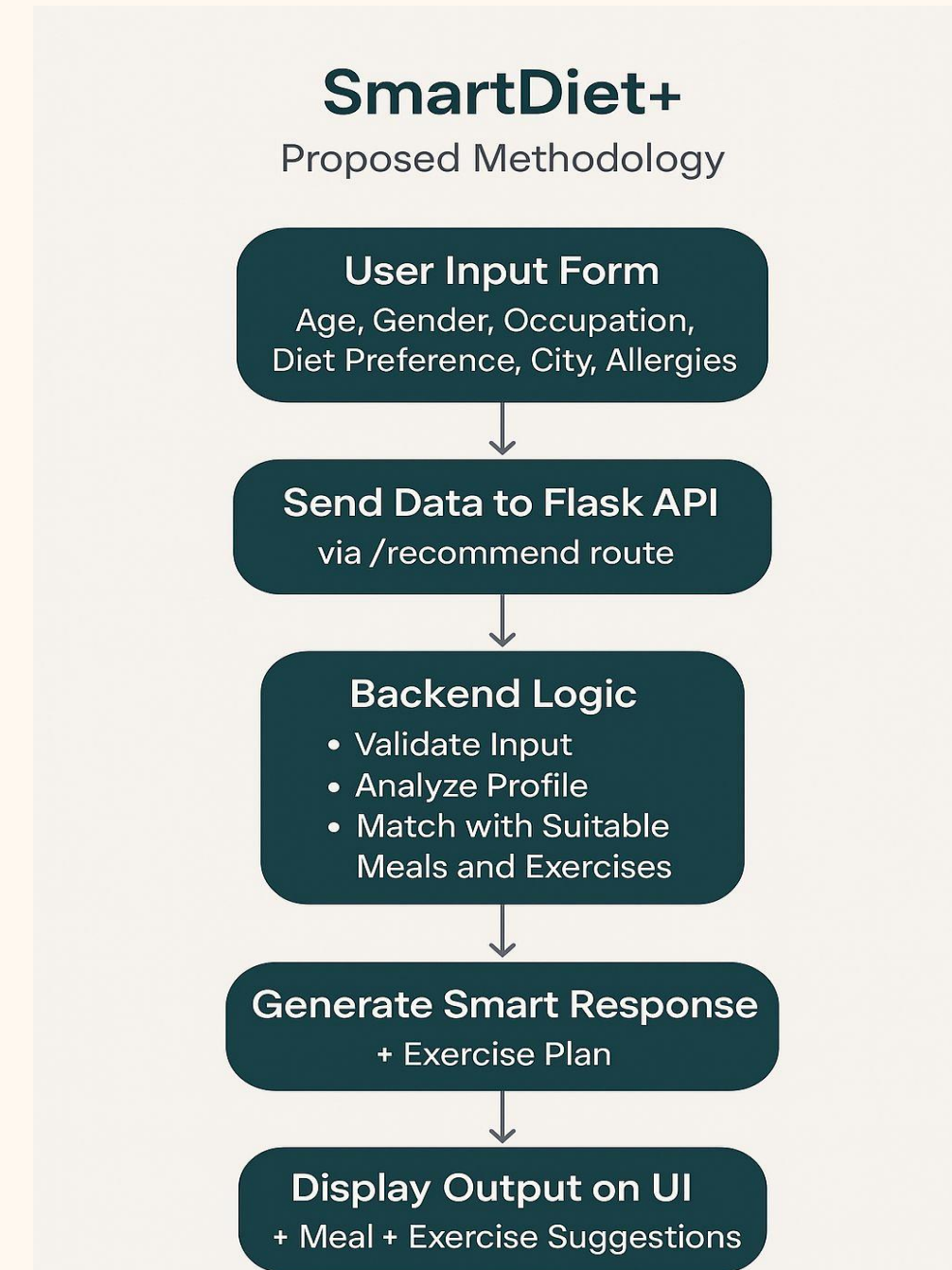


Fig 1

7. Conclusion

The Diet Recommendation System presents a smart, personalized approach to meal planning and calorie tracking, integrating machine learning, AI-driven recommendation models, and automated tracking mechanisms to simplify diet management.

This system goes beyond conventional meal planning by incorporating real-time feedback loops, which enable continuous improvements based on user behavior. The nutrition database integration ensures that dietary recommendations are well-informed and aligned with standard health guidelines. Moreover, adaptive meal planning dynamically adjusts meal suggestions based on the user's dietary preferences, health goals, and past consumption patterns.

By leveraging these advanced features, the system becomes an efficient and intuitive tool for individuals striving to maintain a healthy lifestyle, offering data-driven, personalized, and adaptable meal planning for optimal nutrition and well-being.



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